

Intermediate Macroeconomics: Lecture W5 (Ch. 12)

ECON 311 – Intermediate Macroeconomics

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What are You Going to Learn (Ch.12)

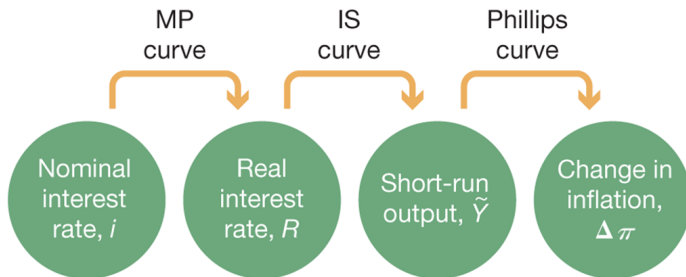
- How the central bank effectively sets the real interest rate in the short run
 - Establish Monetary Policy (MP) curve in our short-run model
- The Phillips curve describes how firms set their prices over time, pinning down the inflation rate
- How the IS curve, the MP curve, and the Phillips curve make up our short-run model
- How to analyze the evolution of the macroeconomy in response to changes in policy or economic shocks

Introduction (Ch.12)

- The central bank has the policy tool to adjust the overnight lending rate between banks
- This interest rate impacts investment decisions, financial markets, and GDP
- In previous chapter, we cover the IS curve, which show how the real interest rate determines output
- This chapter introduce the Monetary Policy (MP) curve and Phillips curve
 - MP curve: show how the central bank determines the nominal interest rate – which can influence the real interest rate
 - Phillip Curve (PC): how changes in output relate with the evolution of inflation

The Structure of the Short-Run Model

- Through the MP curve, the central bank sets the nominal interest rate, which determines the real interest rate
- Through the IS curve, the real interest rate influences short-run GDP
- The Phillips curve describes how the state of the economy influences changes in inflation



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The MP Curve

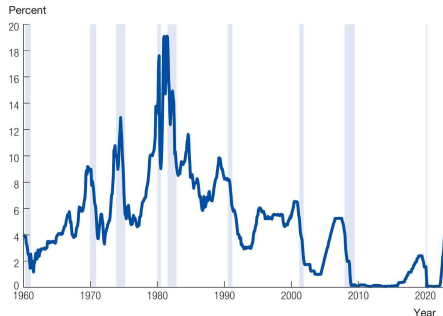
- Most advanced economies use the short-term nominal interest rate as the key policy tool
 - The institution setting this interest rate is the central bank
 - In the US, it is Federal Reserve, in the Canada, it is the Bank of Canada and in the EU, it is European Central Bank (ECB)
- Large banks and financial institutions regularly borrow and lend to one another daily
 - To maintain enough cash or reserve
- The financial institution in the system can borrow or lend with the central bank as well
- Central banks set the nominal interest rate by stating the rate at which they are willing to lend or borrow

The MP Curve

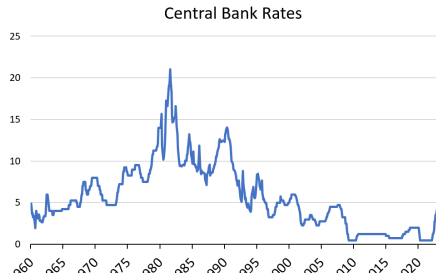
- Banks borrowing from each other must match the central bank lending rate
 - Banks cannot charge a higher rate, everyone would use the central bank
 - Banks cannot charge a lower rate, everyone would borrow at the lower rate and lend it back to the central bank at a higher rate (arbitrage)
- The central bank's willingness to borrow and lend at a specified rate pins down the overnight rate

Central Bank Rates – US and Canada

- there is significant variation in both country, but move closely with each other
- In the 1980s, the fed funds rate reached almost 20 percent
- Also, during the early 2000s, the interest rate fell during and after the 2001 recession and remained low for about 2 years
- During the financial crisis of 2007-09, central bank drop the rate and remain low for abnormal long period, 6+ years



Source: Effective Federal Funds Rate [FEDFUNDS], retrieved from FRED, Federal



From Nominal to Real Interest Rates

- Real interest rate is what affects economic activity and enters the IS curve, no nominal interest rate
- We can use Fisher equation to relate nominal interest rate to real interest rate:

$$i_t = R_t + \pi_t$$

Nominal Interest Rate = Real Interest Rate + Inflation

- Changes in the nominal interest rates affect the real interest rate, so long as they are not offset by corresponding changes to inflation

$$R_t = i_t - \pi_t$$

From Nominal to Real Interest Rates

- Sticky inflation assumption
- In the short run, inflation
 - displays inertia, or stickiness
 - adjusts slowly over time
 - does not respond directly to monetary policy
- This applies in the very short run, that is, 6 months or so
- Central banks can set the real interest rate in the short run

Ex Ante and Ex Post Real Interest Rates

- Sophisticated version of the Fisher equation:

$$i_t = R_t + \pi_t^e$$

where π_t^e is the expected rate of inflation

- In practice, people borrow before the rate of inflation is known. Expected inflation is actually what affects the decision to borrow
- The rate relevant for investment decisions:

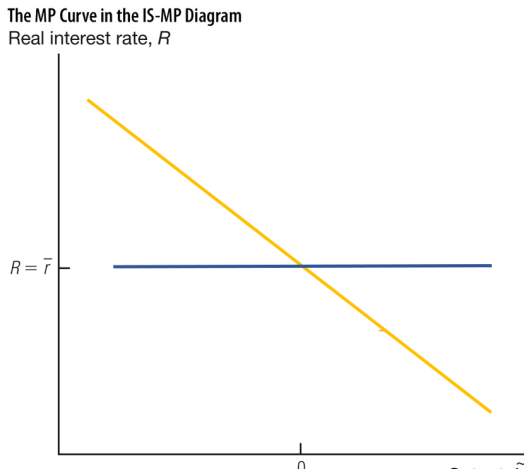
$$R_t^{\text{ex ante}} = i_t - \pi_t^e$$

- The actual real interest rate:

$$R_t^{\text{ex post}} = i_t - \pi_t$$

The IS-MP Diagram

- The MP curve: Illustrates the central bank's ability to set the real interest rate at a particular value (horizontal line)
- The IS curve: Illustrates the negative relationship between interest rates and short-run output

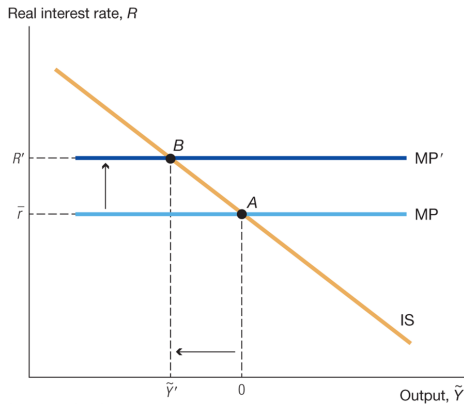


The IS-MP Diagram

- Typically, the economy start at potential. That is, (i) Real interest rate = MPK, (ii) No aggregate demand shocks, (iii) Short-run output = 0
- If the central bank raises the interest rate above the MPK
 1. inflation is sticky (slow to adjust)
 2. the real interest rate rises
 3. investment falls

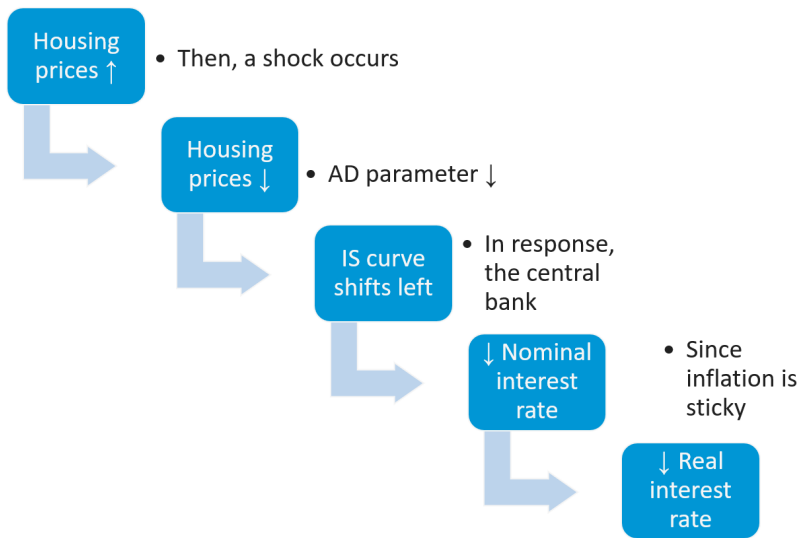
Raising the Interest Rate in the IS-MP Diagram

- Inflation is slow to adjust. So, an increase in the nominal interest rate raises the real interest rate, such that $R > \bar{r}$
- Firms and households reduce investment
- Output declines and the central bank can cause a recession



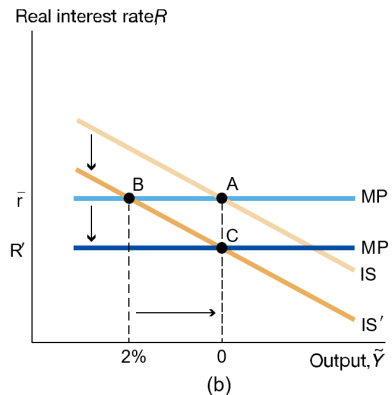
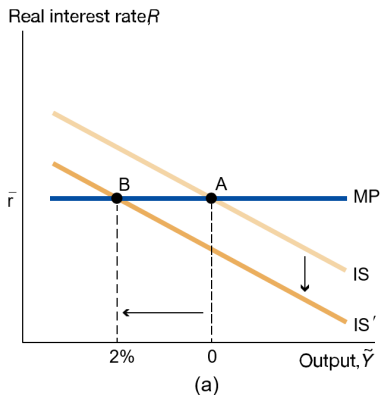
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The End of a Housing Bubble



Stabilizing the Economy after a Housing Bubble

- The negative AD shock moves the economy from point A to point B
- The central bank recognizes the potential for a recession, and stimulates the economy with lower interest rates (shift MP curve down)
- Output moves back to potential at point C



The Phillips Curve

- The inflation rate is the percent change in the overall price level

$$\pi_t = \frac{P_{t+1} - P_t}{P_t}$$

- Firms set their prices on the basis of
 - their expectations of the economy-wide inflation rate (π_t^e)
 - the state of demand for their product ($\bar{\nu} \tilde{Y}_t$)
- Inflation: $\pi_t = \pi_t^e + \bar{\nu} \tilde{Y}_t$
- When the economy is in downturn ($\tilde{Y} < 0$), firms would lower their price to keep the sale (i.e., demand curve shift left)
- Otherwise, the firm would set the price based on what they expect the overall inflation is going to be

The Phillips Curve

- A simple form of the expected inflation can take form of:

$$\pi_t^e = \pi_{t-1}$$

- This form is called the Adaptive Expectations:
 - Firms expect next year's inflation rate to be the same as this year's inflation rate
 - Firms adjust their forecasts of inflation slowly
 - Consistent with our sticky inflation assumption

The Phillips Curve

- The Phillips curve describes how inflation evolves over time as a function of short-run output
- Combine the two equations we had from previous slides

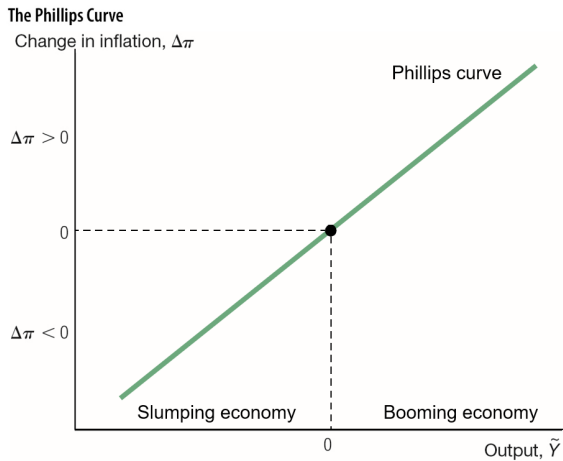
$$\pi_t = \pi_{t-1} + \bar{\nu} \tilde{Y}_t$$

- If output is below (above) potential, prices rise more slowly (rapidly) than usual

$$\Delta\pi_t = \pi_t - \pi_{t-1} = +\bar{\nu} \tilde{Y}_t$$

- The Phillips curve can be expressed as $\Delta\pi_t = \bar{\nu} \tilde{Y}_t$
 - $\bar{\nu}$ indicates the sensitive inflation is to demand conditions

The Phillips Curve



Price Shocks and the Phillips Curve

- To account for temporary increases in the price of inflation, we can add shocks

$$\pi_t = \pi_{t-1} + \bar{\nu} \tilde{Y}_t + \bar{o}$$

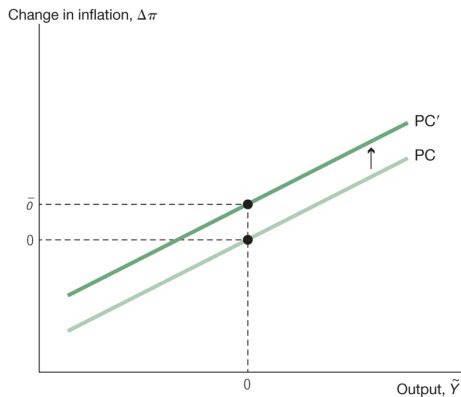
or

$$\Delta\pi_t = \pi_t - \pi_{t-1} = \bar{\nu} \tilde{Y}_t + \bar{o}$$

- The actual rate of inflation now depends on three things:
 - The rate of inflation firms expect (past inflation rate)
 - Demand conditions (adjustment for the state of the economy)
 - Shocks to inflation

Oil Price Shock

- A rise in oil prices has an immediate and highly visible impact on many prices in the economy.
 - It changes the price of gasoline, an airline ticket and heating a home, etc
 - The cost of an increase in oil prices has both direct and indirect effects



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Cost-Push and Demand-Pull Inflation

- Cost-push inflation
 - Price shocks to an input in production
 - Tends to push the inflation rate up
- Demand-pull inflation
 - Changes in short-run output
 - Increases in AD pull up the inflation rate

The Phillips Curve and the Quantity Theory

- The quantity theory: a decrease in growth rate of real GDP would lead to an increase in inflation]
 - According to Milton Friedman, inflation is caused by “too much money chasing too few goods.”
- The Phillips curve: a decrease in growth rate of real GDP would lead to a decrease in inflation (Not increase)

The Phillips Curve and the Quantity Theory

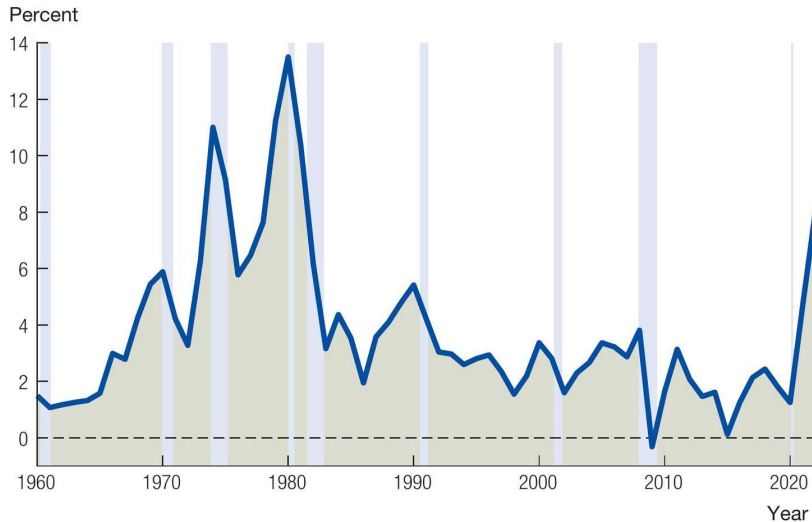
- In the quantity theory, an increase in real GDP reflects an increase in the supply of goods, which lowers prices
 - This is a long-run model and is supply-driven view
- In the Phillips curve, an increase in short-run output reflects an increase in the demand for goods
 - When firms are faced with an increase in demand, they raise their prices
 - This is part of our short-run model and is a demand-driven view

Using the Short-Run Model

- Combine the components of the short-run model to determine output and inflation to illustrate two examples
 - Disinflation: a sustained reduction in inflation to a stable, lower rate
 - This is seen in how the economy moved from a period of high and uncertain inflation in the 1970s to an extended period of more than two decades of low and stable inflation
 - Causes of the Great Inflation of the 1970s
 - This shows how misinterpreting that decade's productivity slowdown contributed to the rising inflation

Inflation in the United States

- In the 70s, the inflation rise rapidly to the highest level since WWII



Source: FRED, Federal Reserve Bank of St. Louis;
fred.stlouisfed.org/series/CPIAUCSL. Recessions are shaded. The inflation rate is computed as the percentage change in the consumer price index.

The Volcker Disinflation

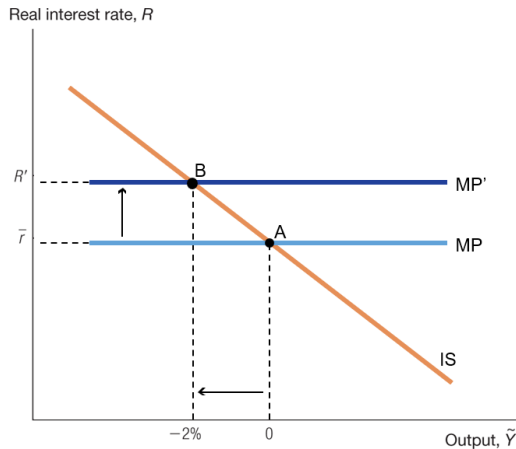
- Paul Volcker was appointed to chair the Federal Reserve Board of Governors in 1979
- Inflation in 1979 was over 10 percent, partially due to
 - loose monetary policy in previous years, and
 - oil shocks in 1974 and 1979
- Volker was supposed to bring inflation back under control

The Volcker Disinflation

- The long-run theory of inflation tells us the money supply must be reduced
 - This requires tight money, or an increase in the nominal interest rate
- However, the classical dichotomy does not hold in the short run
 - Due to sticky inflation, increasing the nominal interest rate will increase the real interest rate
- The real interest rate increase induces a recession
 - The recession causes negative changes in inflation
 - As demand falls, firms raise their prices slower to sell more

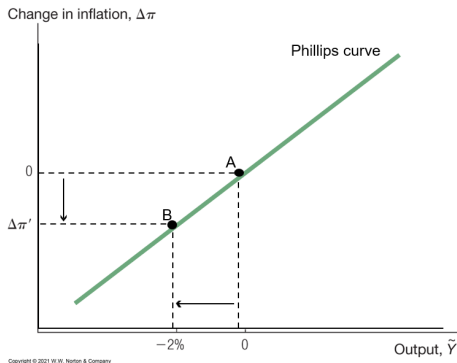
Tightening Monetary Policy

- A rise in the real interest rate causes $R > \bar{r}$ due to sticky inflation
- Firms and households put their investment plans on hold, which causes output to fall from point A to point B. The economy enters a recession



A Recession and Falling Inflation

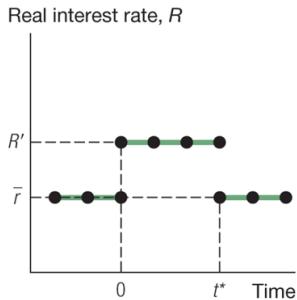
- A recession causes the change in the inflation rate to become negative
- If the fall in inflation is persistent and forms the expectation going forward, the expectation term in PC would fall as well



The Volcker Disinflation

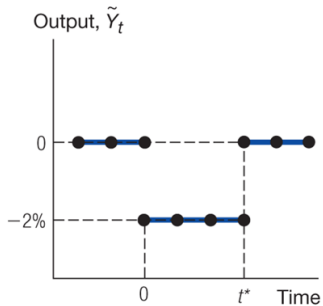
- Volcker-style policy can keep the real interest rate high and output below potential, until inflation falls to a more appropriate level
 - At the cost of a slumping economy
 - High unemployment and lost output
 - In a lot of developing countries, it may cause civil unrest
- Once inflation has declined sufficiently, real interest rate can fall back to MPK, allowing output to rise back to potential

The Disinflation over Time

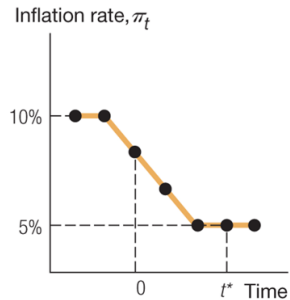


(a) The Fed raises the interest rate...

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(b) causing a recession...



(c) which leads inflation to fall.

The Great Inflation of the 1970s

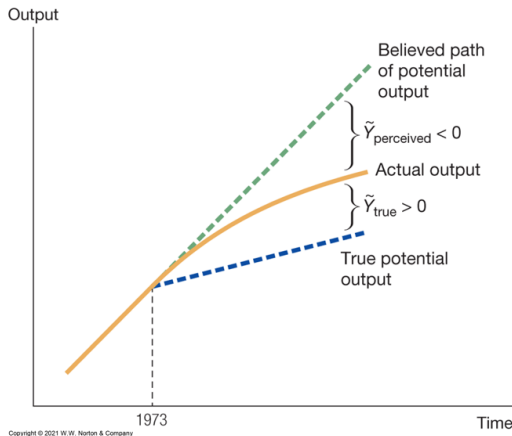
- Inflation in the United States and in many other industrialized countries was relatively low and stable in the 1950s and 1960s
- Inflation rose in the 1970s for three reasons:
 - OPEC coordinated oil price increases: Oil shock
 - U.S. monetary policy was too loose
 - The belief at this time was that there was a permanent trade-off between inflation and unemployment
 - Reducing inflation could only be accomplished by a permanent increase in the rate of unemployment (Paul Samuelson and Robert Solow)
 - However, new theory showed that disinflation required a temporary recession, not a permanent reduction in output (Milton Friedman, Edmund Phelps, Robert Lucas)

The Great Inflation of the 1970s

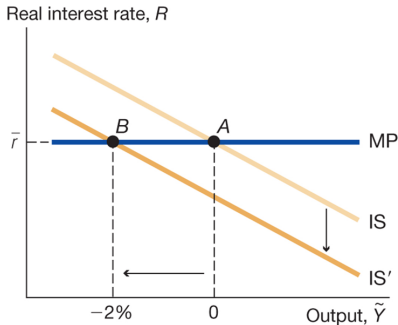
- Inflation rose in the 1970s for three reasons:
 - The Federal Reserve did not have perfect information
 - There was a productivity slowdown that started in the early 1970s ($\bar{Y}$ grow slower than before)
 - However, this was interpreted as a shock to short-run output ($\tilde{Y}$), pushing it below zero
 - Policymakers reduced interest rates to stimulate the economy, which led to higher inflation

Mistaking a Slowdown in Potential for a Recession

- The policymaker thought the short-run output was negative and decided to lower the interest rate
- Meanwhile, it may boost short-term output. However, in the long run, the economy experienced a period of high inflation

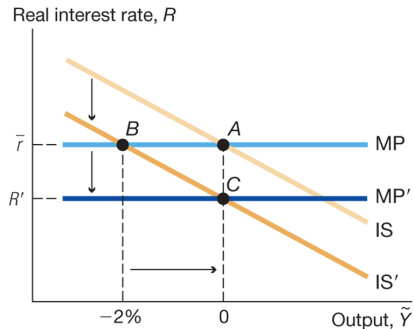


The Great Inflation



(a) Policymakers believe the slow-down in growth is caused by a negative aggregate demand shock...

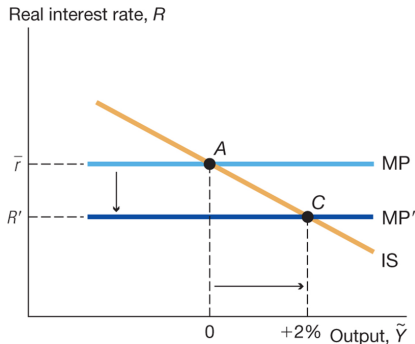
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(b) so they lower R .

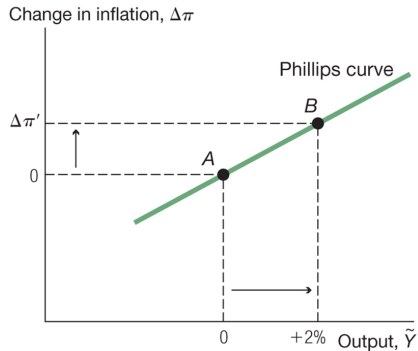
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The Great Inflation



(c) In truth, the IS curve never shifted...

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(d) so $\downarrow R \Rightarrow \uparrow \tilde{Y} \Rightarrow \uparrow \pi$.

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Microfoundations: Understanding Sticky Inflation

- Sticky inflation is one of the essential building block of the MP curve, which imply changes in the nominal interest rate affect the real interest rate because inflation does not adjust immediately
- This is also critical for the Phillips curve, the assumption of adaptive expectations assumes that expectations of inflation adjust slowly over time

- The classical dichotomy states that changes in nominal variables have only nominal effects
 - The real side of the economy is determined solely by real forces
 - If the central bank decides to double the money supply, then all prices can double
 - If monetary policy affects real variables, the classical dichotomy fails in the short run
 - Explaining this failure is one of the crucial requirements of a good short-run model

The Classical Dichotomy in the Short Run

- Can the classical dichotomy hold at all points in time?
 - All prices, including wages and rental prices, must adjust in the same proportion immediately
- However, this most likely not happening. Reasons that the classical dichotomy fails in the short run:
 - Imperfect info
 - Firms are constantly being hit by a wide range of individual-specific shocks
 - Monetary policy may not be one of the most important shocks to consider
 - Subtle monetary policy changes make price coordination difficult
 - Also, monetary policy changes occur through the interest rate, not through inflation directly, so implementation also depends on other factors

The Classical Dichotomy in the Short Run

- Reasons that the classical dichotomy fails in the short run:
 - Costs of setting prices
 - It is too costly and time-consuming to gather information on every detail affecting a firm, and to figure out the correct price every single day
 - Contracts
 - In the short run, contracts set prices and wages in nominal rather than real terms
 - Bargaining costs
 - Negotiating over prices can be risky for job security
 - In addition, it is costly and difficult and inefficient on a daily basis

The Classical Dichotomy in the Short Run

- Reasons that the classical dichotomy fails in the short run:
 - Social norms
 - Money illusion refers to the fact that people sometimes focus on nominal rather than real magnitudes
 - People may feel more strongly about nominal wages, regardless of the overall price level in the economy

Microfoundations: How Central Banks Control Nominal Interest Rates

- In the short-run model, we assume the central bank “sets” the nominal interest rate
- The central bank controls the level of the nominal interest rate by supplying the money that is demanded at that rate
- Develop a simple money demand and supply model to illustrate that
 - Start with the case when central bank control the money supply

Money Supply and Demand

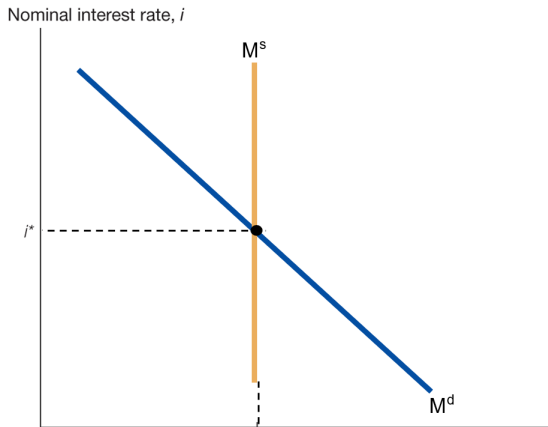
- The nominal interest rate is the opportunity cost of holding money
 - The amount you give up by holding the money instead of keeping it in a savings account
 - Assume you have \$100 remaining after covering your current consumption. You have the option to either retain it as cash or deposit it into a savings account
 - If the interest rate is 1%, the difference is negligible. However, with an interest rate of 20%, you would aim to hold as little cash as possible
- The price in this market is the interest rate

Money Supply and Demand

- The demand for money: decreasing function of the nominal interest rate (i.e, downward sloping)
- The supply of money determined by the level of money the central bank provides and is vertical
- The nominal interest rate is pinned down by the equilibrium in the money market, where households are willing to hold just the amount of currency that the central bank supplies

Money Supply and Demand

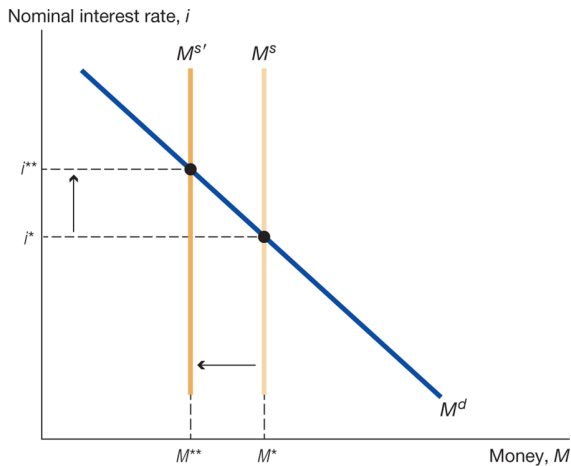
- If the nominal interest rate i is higher than i^* , then households would want to hold their wealth in savings accounts rather than currency, so money supply would exceed money demand
 - This puts pressure on the nominal interest rate to fall
- If i is lower than i^* , money demand would exceed money supply, leading i to rise



Changing the Interest Rate

- To increase interest rates:
 - The central bank reduces the money supply, which creates an excess of demand over supply
 - A higher interest rate on savings accounts reduces excess demand
 - The markets adjust to a new equilibrium

Raising the Nominal Interest Rate



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Why i_t instead of M_t ?

- Historically, central banks did in fact focus directly on the money supply
 - Both in the US and much of Europe in the 1970s and early 1980s
- The interest rate is crucial even when central banks focus on the money supply
 - After all, that is the variables affecting economic activity
- The money demand curve is subject to many shocks, which shift the curve
 - Changes in price and output (higher demand for good require more cash in advance)
 - Changes in technology (e.g., advent of ATMs, the increasing use of credit and debit cards, and the increased availability of financial products like mutual funds and money market accounts)
 - In the 80s, economist found that it is very hard to pinpoint the demand curve

Why i_t instead of M_t ?

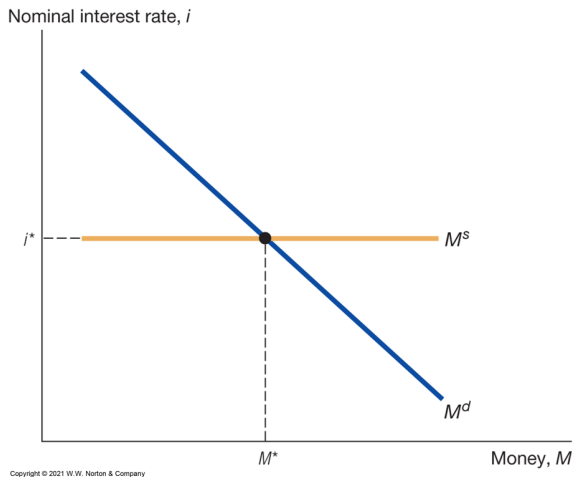
- With a constant money supply, the nominal interest rate would fluctuate, leading to changes in the real interest rate and changes in output if the central bank did not act
- By targeting the interest rate directly, the central bank automatically adopts a policy that will adjust the money supply to accommodate shocks to money demand

Why i_t instead of M_t ?

- Such a policy prevents the shocks from having a significant effect on output and inflation and helps the central bank stabilize the economy
- An expansionary (loosening) monetary policy: Increases the money supply or Lowers the nominal interest rate
- A contractionary (tightening) monetary policy: Reduces the money supply or Increases the nominal interest rate

Why i_t instead of M_t ?

- The money supply schedule is effectively horizontal at a targeted interest rate
 - When the central bank sets an interest rate, it is willing to supply whatever amount of money is demanded at the interest rate



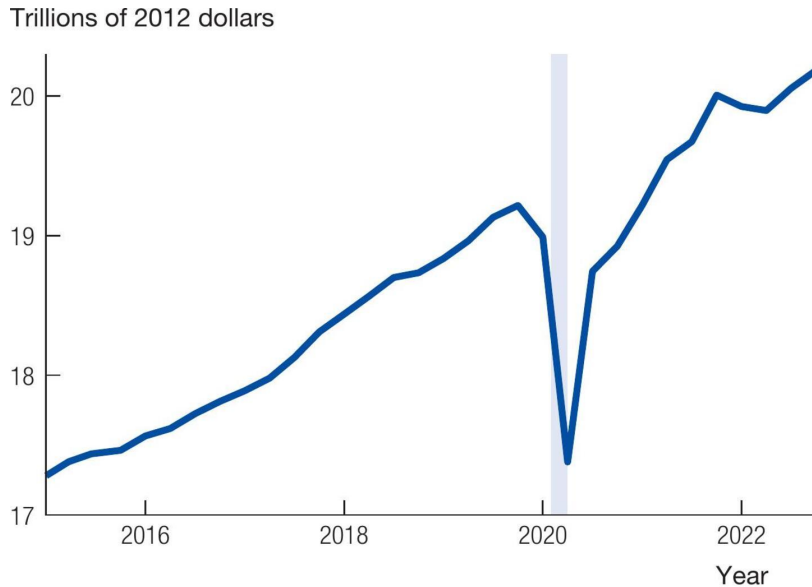
The Covid-19 Recession

Covid-19 Recession

- GDP sharply declined by 10 percent in 2020
- By 2022, GDP had returned to trend
- Inflation declined somewhat in 2020 at the beginning of the recession
- By 2021, inflation started rising and continued to levels not seen since the 1970s

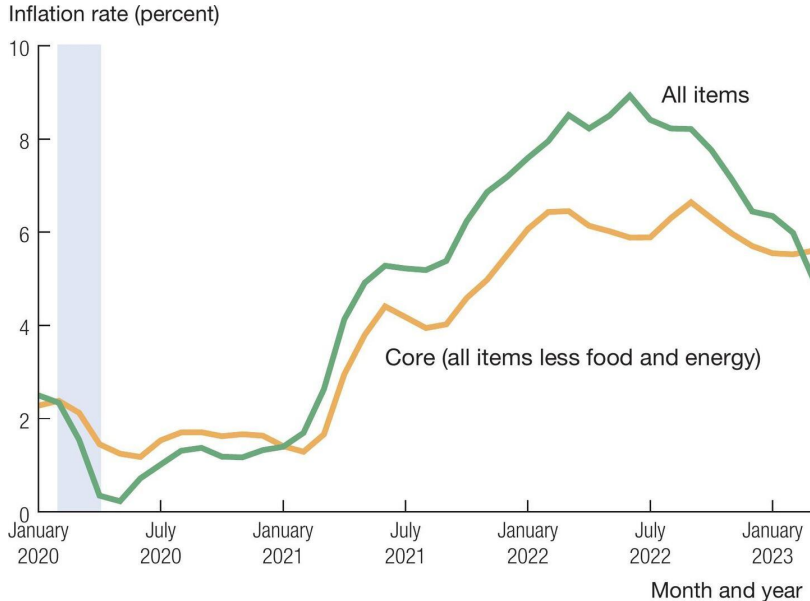
Why did GDP fall sharply and then quickly rebound? Why did inflation initially decline in 2020, and then rise quickly in 2021 and 2022?

Covid-19 Recession and Real GDP in the United States



Source: FRED, Federal Reserve Bank of St. Louis; fred.stlouisfed.org/series/GDPC1

Covid-19 Recession and Consumer Price Index Inflation in the U.S.



Source: FRED, Federal Reserve Bank of St. Louis, series CPIAUCSL and CPILFESL.

Covid-19 as an Aggregate Demand Shock

Covid-19 acted as a “tax” on consumption

- Possibility of contracting Covid-19 reduced in-person retail, in-person dining, travel, and in-person entertainment activity
- Consumption fell sharply

$$\bar{a}_C \downarrow \rightarrow \bar{a} \downarrow = \tilde{Y}_t \downarrow \rightarrow \pi_t \downarrow$$

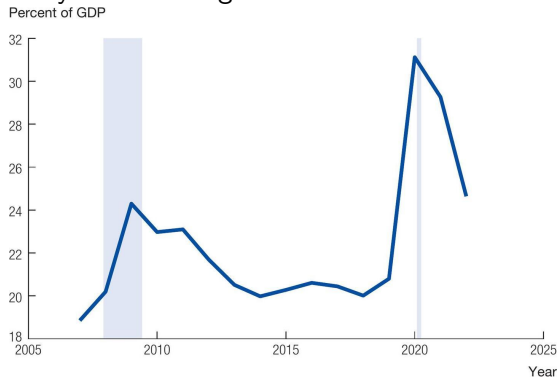
Covid-19 as a Shock to $\bar{Y}$

Covid-19 acted as a “tax” on working

- Possibility of contracting Covid-19 if you left home to go to work reduced employment both voluntarily and involuntarily
- Employment declined sharply, reducing production
- $\bar{Y}$ declines

U.S. Federal Government Spending as a Share of GDP

- The sharp increase in government spending during the Covid-19 pandemic buffered the economy from negative economic impacts
- Although unemployment reached levels not seen since the Great Depression, household incomes increased for many because of government intervention



Source: FRED, Federal Reserve Bank of St. Louis, series FYONGDA188S.

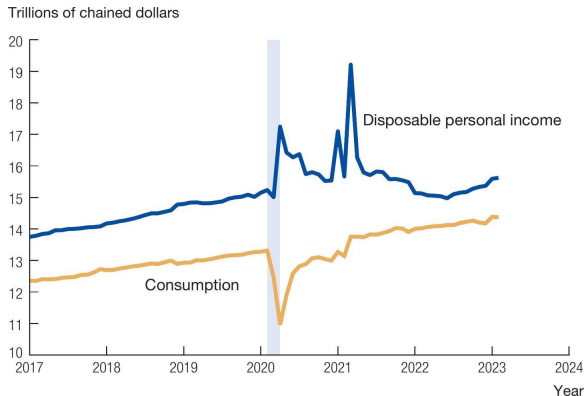
- Resulting an increase in $\bar{a}_\sigma$

Another reason for the strong economic recovery was government policy

- Government spending as share of GDP increased from 22 percent in 2019 to 32 percent in 2020
- Loans, unemployment benefits, and direct transfers to businesses and individuals allowed for continued consumption
- Disposable personal income increased from 2020 to 2021 despite a 15 percent unemployment rate

Income and Consumption

- The three spikes in disposable personal income correspond to three separate direct payments from government to households
- The simultaneous increase in disposable personal income and decrease in consumption resulted in increased household savings and a decline in poverty



Source: FRED, Federal Reserve Bank of St. Louis, series DSPIC96 and PCEC96.

Central Bank's Responses

- Bank of Canada lowered the bank rate from 2% to close to 0%
- The MP curve shift down

Combined Shock

- The economy was shocked with
 - a decrease in $\bar{a}_c$
 - a decrease in $\bar{Y}$
- The government and central bank responses with:
 - an increase in $\bar{a}_g$
 - a decrease in R_t
- In combination, the economy experienced
 - A large fall in current output Y_t
 - Mild decrease in $\tilde{Y}_t$ as the decrease in $\bar{Y}$ and government/central bank helps lowered the impact
 - Resulting a mild decrease in $\Delta\pi_t$

The Rapid Recovery of GDP

- Production can recover quickly from negative shocks
 - Happens after every weekend
 - In Europe, many manufacturers shut down in August for vacation and restart in September
- The Covid-19 recession was short-lived because people quickly returned to normal economic activity
 - Improved scientific understanding of the virus
 - Widespread availability of masks
 - Effective vaccines

The Return of Inflation

- After decades of 2 percent inflation, the increase in inflation rates in 2021–2022 surprised most observers
- How did this happen?
 - Covid-19 pandemic interfered with global supply chains
 - Energy price shocks associated with the war in Ukraine
 - Demand for goods outpaced production

Next Lecture

- Next lecture will cover Ch.13